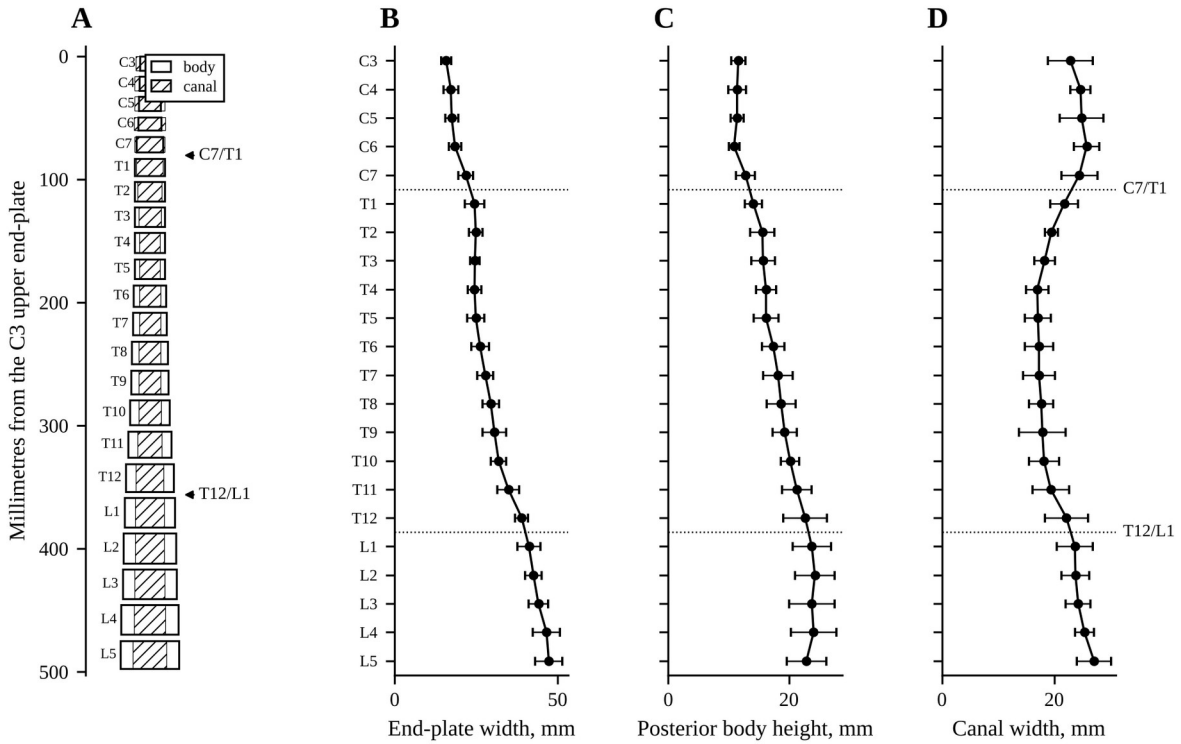
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The junction, named from the shape profile alone



Panjabi's tables give every level a mean and a spread for the end-plate, the body and the canal; drawn to their millimetres, the column is the shape a reader of CT sees

The cervicothoracic step is large in every measure; the thoracolumbar step is small beside the spread of one level

A window of bodies is named by scoring every naming the column allows against this profile, with no count from either end

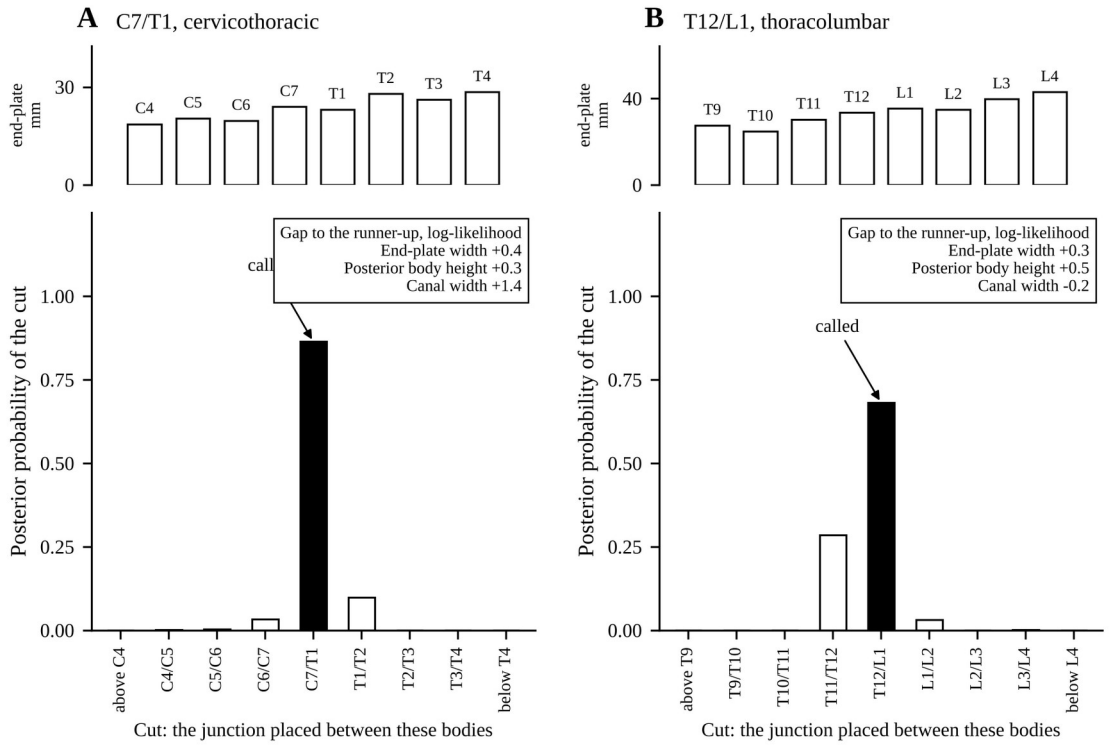
Panjabi MM, et al. Spine 1991;16:861 and 888; Spine 1992;17:299. Published morphometry drawn to scale; the bars are one population SD. No patient data.

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How it decides: every naming scored, the gap explained



Every naming a column with 11 to 13 thoracic and 4 to 6 lumbar bodies allows is scored per measure; a free offset per measure cancels the patient's size and the measurement convention

The posterior over the cut, averaged over the namings that share it under a flat prior, is the answer; the per-measure gap to the runner-up says why: at C7/T1 the canal width decides, at T12/L1 no measure decides strongly

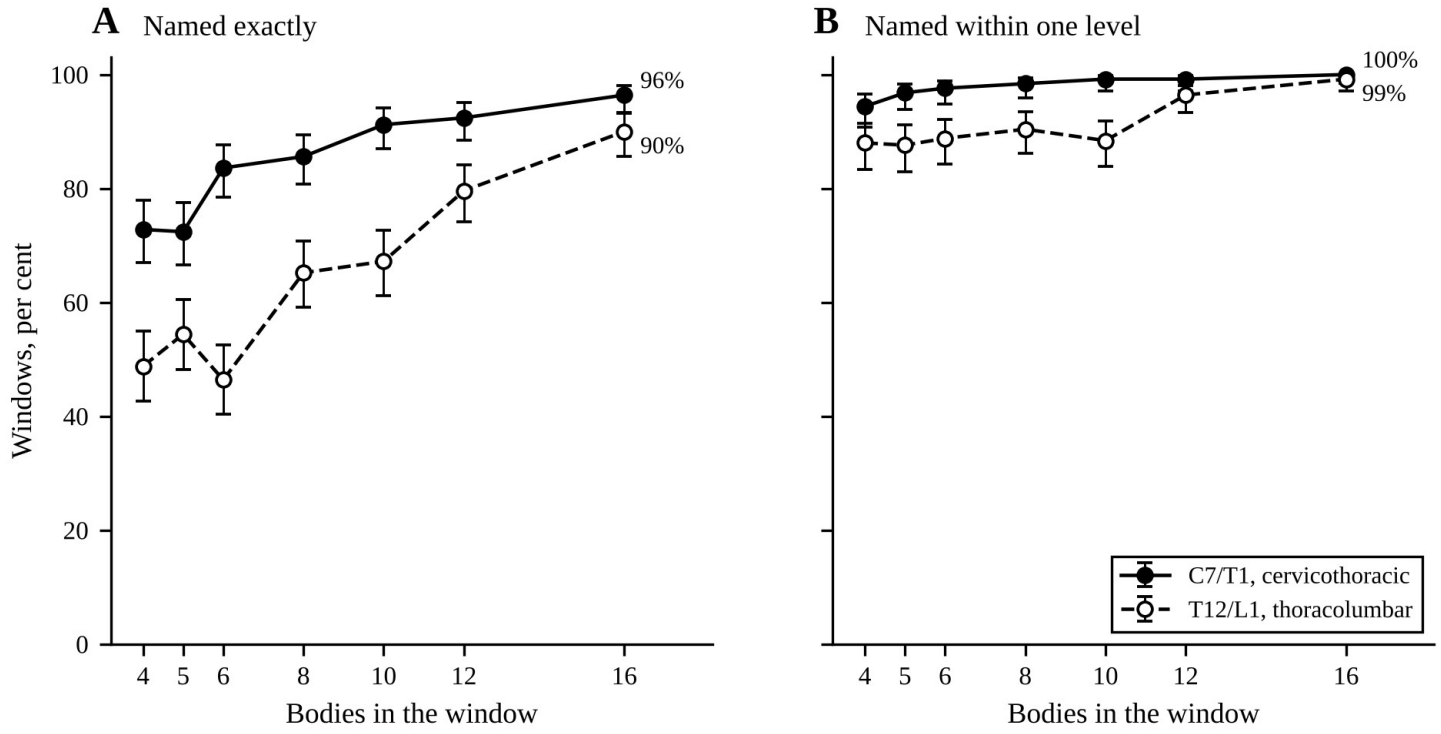
A margin below a chosen floor is an abstention, not a guess

One synthetic column from the tables, eight bodies around each junction; Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

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Result: C7/T1 named robustly, T12/L1 the hard junction



Eight bodies in view: C7/T1 named exactly in 86 of 100 windows and within one level in 98; T12/L1 exactly in 65, within one in 90

Sixteen bodies: 96 and 90 exactly. Four bodies: 73 and 49

Fitted on no CT: the model is read off the published tables, so every number is about the tables and the generator, not yet about a patient

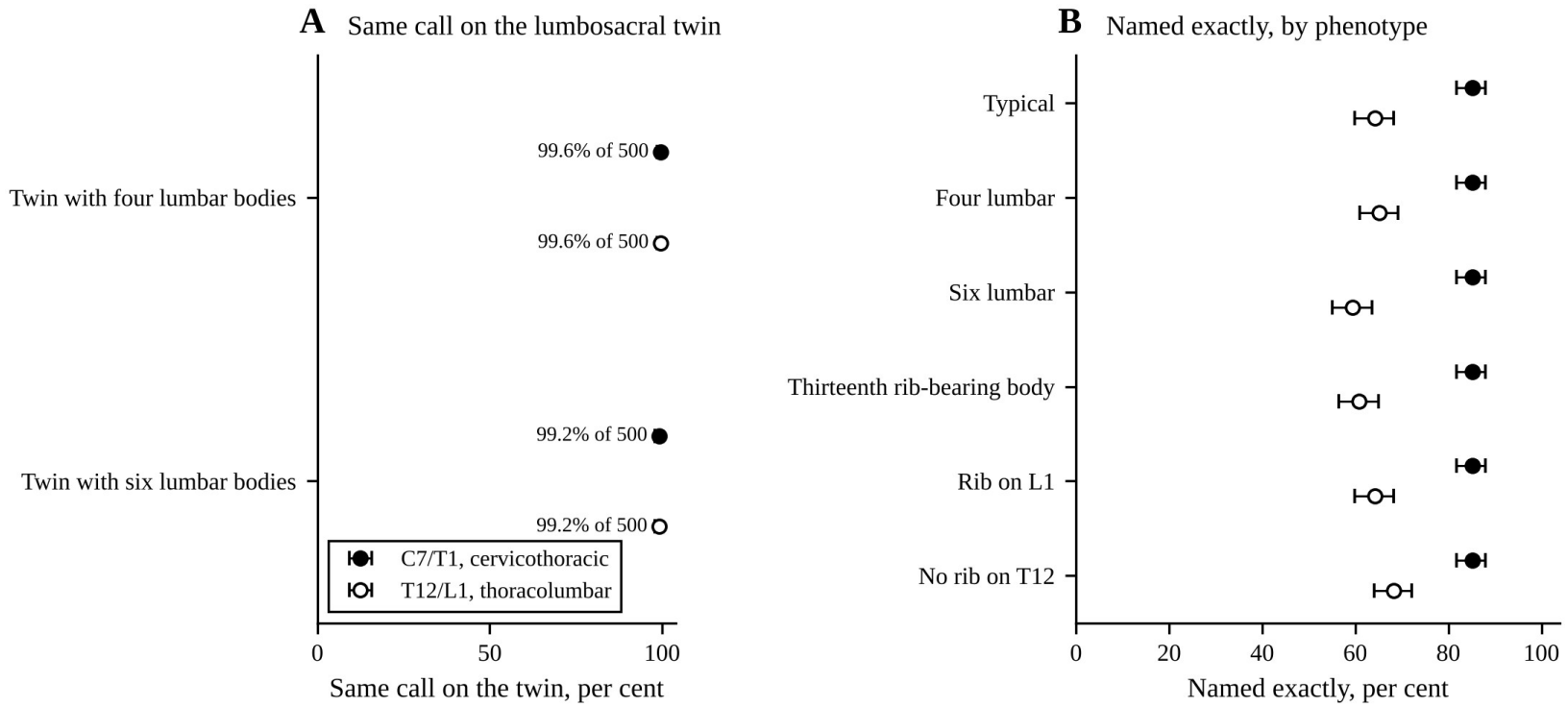
Profile template, 250 synthetic typical windows per width, measures end-plate width, posterior body height, canal width; 95% Wilson intervals. Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

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A lumbosacral anomaly does not move the junction call



One column drawn once and named two ways, four or six lumbar bodies: the C7/T1 call is unchanged in 99.6 and 99.2 of 100 twins; T12/L1 in 99.6 and 99.2

By phenotype, C7/T1 holds at 85 to 85% whatever the column; T12/L1 runs from 59% with six lumbar bodies to 68% with no rib on T12, 64% on a typical column

The lumbosacral count lies outside the junction's window; the transitional body lies inside it (next slide)

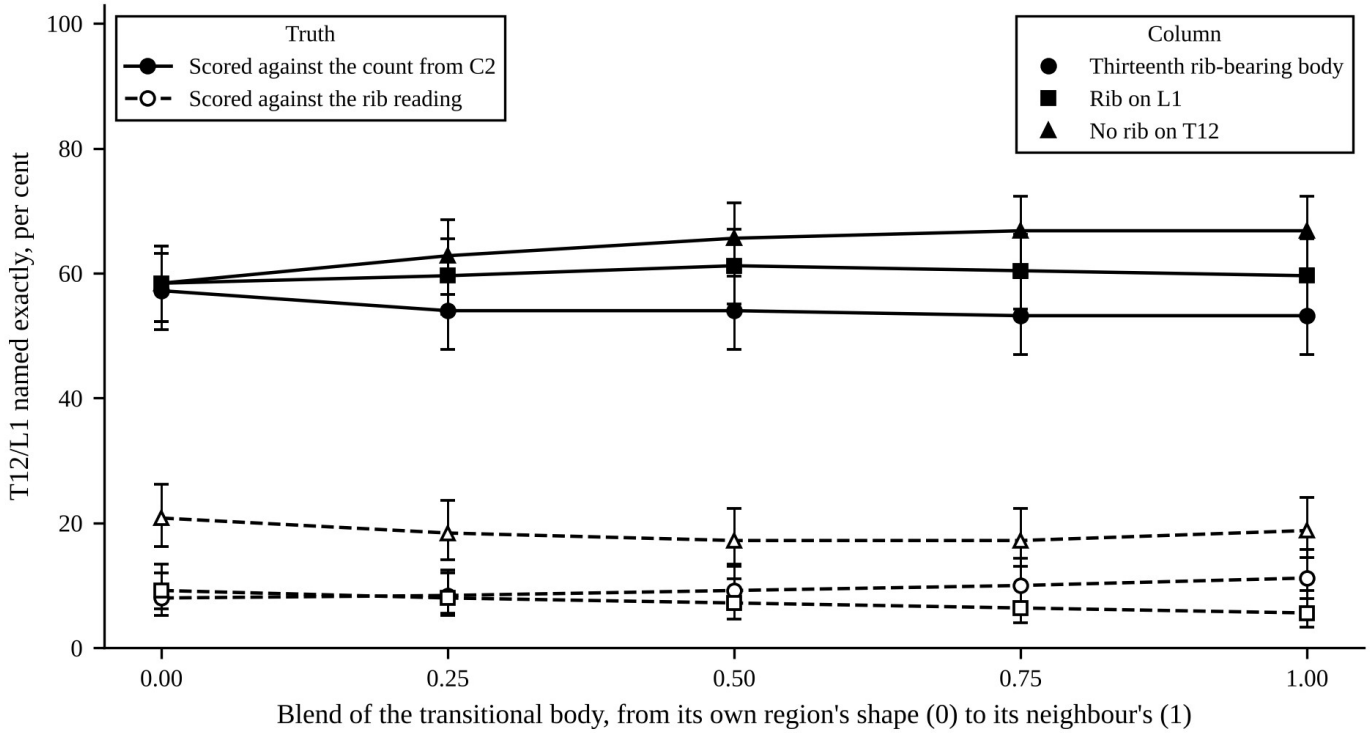
Profile template, 500 synthetic windows per class, measures end-plate width, posterior body height, canal width; 95% Wilson intervals. Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

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The transitional body: two truths one level apart



Blending the transitional body toward its neighbour's shape leaves T12/L1 naming by the count from C2 at 53 to 67% across the three phenotypes, and within one level at 83 to 92%

Scored against the ribs instead, the same calls are exact in only 6 to 21%: a rib-bearing L1 and a thirteenth thoracic body are one shape, and only the count from C2 tells them apart

That is the case a full-spine corpus counted from C2 meets, with the ribs read independently

Profile template, 250 synthetic windows per point; solid, scored against the count; dashed, against the ribs. Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026.

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What is validated, what is not, and what comes next

Column	C7/T1 exact	C7/T1 within one	T12/L1 exact	T12/L1 within one
Typical	85% (426/500)	97% (487/500)	64% (321/500)	91% (453/500)
Four lumbar	85% (426/500)	97% (487/500)	65% (326/500)	89% (445/500)
Six lumbar	85% (426/500)	97% (487/500)	59% (297/500)	91% (457/500)
Thirteenth rib-bearing body	85% (426/500)	97% (487/500)	61% (304/500)	88% (439/500)
Rib on L1	85% (426/500)	97% (487/500)	64% (321/500)	89% (447/500)
No rib on T12	85% (426/500)	97% (487/500)	68% (341/500)	92% (459/500)

Named exactly and within one level, by phenotype, 500 synthetic windows per class; the denominators are the windows with the junction in view. A thirteenth rib-bearing body and a rib on L1 are one anatomy drawn from the two ends of the blend.

Validated: the lumbar measurements against published norms. Not validated: the same measures at cervical and thoracic levels, until each is checked against a published series there

Every number on these slides is from synthetic columns drawn from Panjabi's tables; none is from a patient

Next: full-spine CT counted from C2, every column's class with its denominator; where the shape and the count disagree, the ribs decide, never the model's own call, and the circular arm is shown beside it for scale

The thoracolumbar junction is the one still owed

Synthetic columns drawn from Panjabi's published means and spreads (Spine 1991;16:861, 888; 1992;17:299); no patient data. Retrospective modelling, 2026. Evidence status: modelling on published tables; no clinical use.